

Experiment 3 Guide (C++ Programming)

I. Experimental Objectives

This experiment aims to practice C++ functions, which is chapter 6 of the textbook. The detailed goals of this experiment are as follows.

- **Modularity:** Understand the benefits of dividing a program into smaller, manageable tasks using functions.
- **Function Elements:** Master the three essential parts: Function Declaration (Prototype), Definition, and Call.
- **Data Exchange:** Understand the mechanism of passing parameters and returning values.
- **Library vs. User-defined:** Utilize C++ standard library functions (math, cmath) and create custom user-defined functions.
- **Scope & Lifetime:** Distinguish between local and global variables and understand their visibility in memory.

II. Required Software Tools

Visual Studio code and a C++ compiler

III. Experimental Content

Complete the following 8 programming tasks. Ensure all code includes proper documentation and follows the specific technical requirements for each problem.

IV. Practical Tasks

Task 1: Greeting Function (Void, No Parameters)

- **Goal:** Practice defining and calling a void function with no parameters.
- **Problem:** Write a function `displayHeader()` that prints your name and lab info in a decorative box. Call it at the beginning of your `main()`.
- **Requirements :** Function must be void, No parameters
- **Input Example**

(No user input required)

- **Output Example**

```
*****
* Name: John Doe                *
* Lab: C++ Programming Lab      *
*****
```

Task 2: Square and Cube (Data-Returning, One Parameter)

- **Goal:** Practice writing a data-returning function with parameters.
- **Problem:** Create a function long calculatePower(int base, int exp) that returns $base^{exp}$.
Use this function to compute the square and cube of a user-input number.
- **Requirement:** 1. Define calculatePower(int base, int exp) with return type long.
2. Use the function to calculate both x^2 and x^3 for the input number.
- **Input Example**

Enter a number: 3

- **Output Example**

Square: 9
Cube: 27

Task 3: Maximum of Three (Comparison Logic)

- **Goal:** Practice using a function to organize comparison logic.
- **Problem:** Write a function int findMax(int n1, int n2, int n3) that returns the largest of the three integers.
- **Requirement:** Call findMax() from main() and display the returned maximum value.
- **Input Example**

Enter three numbers: 10 25 7

- **Output Example**

Maximum value: 25

Task 4: Circle Area Calculator (Constants & Math)

- **Goal:** Practice constants, functions, and formatted floating-point output.

- **Problem:** Write a function `double getCircleArea(double radius)` that returns the area of a circle using $A = \pi r^2$.
- **Requirement:** 1. Use constant `PI = 3.14159`. 2. Use `fixed` and `setprecision(2)` for output.
- **Input Example**

Enter radius: 5

- **Output Example**

Circle area: 78.54

Task 5: Temperature Converter (Multiple Functions)

- **Goal:** Practice designing multiple functions in one program.
- **Problem:** Create two functions: `celsiusToFahrenheit()` and `fahrenheitToCelsius()`. Allow the user to choose a conversion type and enter a value.
- **Requirements:** 1. Use `double` for temperature values. 2. Use `fixed` and `setprecision(2)` for output.
- **Input Example 1**

Select conversion:

1. Celsius to Fahrenheit
2. Fahrenheit to Celsius

Enter choice: 1

Enter temperature: 36.5

- **Output Example 1**

Temperature in Fahrenheit: 97.70

- **Input Example 2**

Select conversion:

1. Celsius to Fahrenheit
2. Fahrenheit to Celsius

Enter choice: 2

Enter temperature: 98.6

- **Output Example 2**

Temperature in Celsius: 37.00

Task 6: Prime Number Checker (Boolean Function)

- **Goal:** Practice using a boolean function.
- **Problem:** Write a function `bool isPrime(int n)` that returns true if a number is prime and false otherwise. Use this in `main()` to check a user's input.
- **Requirement:** The input number should be an integer greater than 1. If the input is less than or equal to 1, output "Invalid input."

- **Input Example 1**

Enter a number: 17

- **Output Example 1**

17 is a prime number.

- **Input Example 2**

Enter a number: 18

- **Output Example 2**

18 is not a prime number.

- **Input Example 3**

Enter a number: 1

- **Output Example 3**

Invalid input.

Task 7: Mathematical Library Exploration (cmath)

- **Goal:** Practice using standard library functions from `<cmath>`.

- **Problem:** Write a program that uses `floor()`, `ceil()`, `abs()`, and `log10()` to process a floating-point number.
- **Requirement:** 1. Assume the input is positive. 2. Use `fixed` and `setprecision(2)` for output.
- **Input Example**

Enter a number: 12.75

- **Output Example**

Floor: 12.00
Ceil: 13.00
Absolute value: 12.75
Log10: 1.11

Task 8: Simple Menu System (Modular Program)

- **Goal:** Practice building a modular menu-driven program.
- **Problem:** Create a standalone program with a menu that allows users to choose between Power Calculation, Circle Area, or Prime Check. Each choice should call the corresponding function inside this Task 8 program.
- **Requirement:** Use a menu selection in `main()` and call the appropriate function based on the user's choice. Include a default case or `if-else` branch for invalid menu choices.
- **Input Example 1**

Menu:
1. Power Calculation
2. Circle Area
3. Prime Check
Enter choice: 3
Enter number: 11

- **Output Example 1**

11 is a prime number.

- **Input Example 2**

Menu:

1. Power Calculation
2. Circle Area
3. Prime Check

Enter choice: 1

Enter number: 4

• Output Example 2

Square: 16

Cube: 64

• Input Example 3

Menu:

1. Power Calculation
2. Circle Area
3. Prime Check

Enter choice: 2

Enter radius: 5

• Output Example 3

Circle area: 78.54

• Input Example 4

Menu:

1. Power Calculation
2. Circle Area
3. Prime Check

Enter choice: 5

• Output Example 4

Invalid choice.

V. Submission Requirements

- Submit one zipped folder containing the Lab Report and all C++ source code files.

- Name each source code file using the experiment number and task number, for example:
Experiment_3_Task_1.cpp, Experiment_3_Task_2.cpp, ..., Experiment_3_Task_8.cpp.
- Use the provided Experiment Report Template to complete the report.
- Submit the zipped folder containing the lab report and all C++ source code files to the assignment submission platform before 23:00 on the day after the experiment class:
<https://grader.hitsz.edu.cn/login?next=/courses>